

Nikos Pagonas

Contact Email: nikospagonas00@gmail.com — Website: nikpag.github.io

Education **Columbia University** Sep 2024 – Present
PhD in Computer Science New York, NY, US
Advisor: Prof. [Kostis Kaffes](#)
GPA: 3.94/4.00
Member of [DAPLab](#), focusing on improving the performance and efficiency of agentic serving. I build systems that exploit workflow structure and runtime characteristics to optimize model routing, scheduling, and resource allocation.

National Technical University of Athens (NTUA) Oct 2018 – Jun 2024
Master of Electrical and Computer Engineering Athens, Greece
Advisors: Prof. [Nikos Vasilakis](#), Prof. [Georgios Goumas](#)
GPA: 9.39/10.00 (14th out of 344 students)
Thesis: “*SPLASH: Scaling Out Shell Scripts on Serverless Platforms*”

Experience **Microsoft Research** May 2025 – Aug 2025
Research Intern Redmond, WA, US
Supervisor: [Sharad Agarwal](#)

Google May 2025 – Aug 2025
Student Researcher Sunnyvale, CA, US
Supervisor: Prof. [Arvind Krishnamurthy](#)
Designed and implemented [Cortex](#), a prototype agentic serving platform that leverages workflow awareness and stage isolation to improve scheduling and KV cache efficiency in agentic NL2SQL workflows.

Brown University Sep 2023 – May 2024
Student Researcher Providence, RI, US
Advisor: Prof. [Nikos Vasilakis](#)
Contributed to [PASH](#), an open-source [Linux Foundation](#) project. Worked on improving the performance of shell scripts by running them on serverless infrastructure, and providing fault tolerance for distributed shell script execution.

Arrikto Inc. Dec 2022 – Mar 2023
Software Engineering Intern Athens, Greece
Supervisor: [Vangelis Koukis](#)
Explored how to serve PyTorch models more efficiently through KServe, a highly scalable machine learning deployment platform for Kubernetes.

Awards	Columbia University Presidential Fellowship	2024 – Present
	A.G. Leventis Foundation Scholarship	2025 – Present
	Gerondelis Foundation Scholarship	2025
	ACM SOSP 2025 Student Travel Grant	2025
Publications	VineLM: Trie-Based Fine-Grained Control for Agentic Workflows Nikos Pagonas, Matthew Lou, Tianyi Peng, Dan Rubenstein, Kostis Kaffes	
	Agentic Data Environments Elaine Ang, Chenxi Huang, Georgios Liargkovas, Jerry Liu, Jinhui Liu, Nikos Pagonas, Charlie Summers, Haonan Wang, Jiakai Xu, Tianle Zhou, Yusen Zhang, Zhou Yu, Zhuo Zhang, Tianyi Peng, Kostis Kaffes, Eugene Wu <i>IEEE Data Engineering Bulletin, March 2026</i>	
	AgentOpt v0.1 Technical Report: Client-Side Optimization for LLM-Based Agents Wenyue Hua, Sripad Karne, Qian Xie, Armaan Agrawal, Nikos Pagonas, Kostis Kaffes, Tianyi Peng	
	Cortex: Workflow-Aware Resource Pooling and Scheduling for Agentic Serving Nikos Pagonas, Yeounoh Chung, Kostis Kaffes, Arvind Krishnamurthy <i>SOSP 2025 Workshop on Systems for Agentic AI (SAA 2025)</i>	
	Which Model for Which Role? Sample-Efficient Agent Configuration Search <i>(Under Submission)</i>	
	λ easeh: Scaling Out Shell Scripts with Reliable Cloud Computing <i>(Under Submission)</i>	
Service	Artifact Evaluation Committees EuroSys 2027 , SOSP 2026 , CAIS 2026 , EuroSys 2026 , SOSP 2025 , OSDI 2025 , ATC 2025	
	Web and Publication Chair EuroSec 2024	
Teaching	Columbia University <i>Teaching Assistant</i> Course: Networks, Crowds, and the Web Professor: Augustin Chaintreau	Fall 2025 New York, NY, US
	Columbia University <i>Teaching Assistant</i> Course: Agentic Systems Made Real Professors: Eugene Wu , Kostis Kaffes	Spring 2025 New York, NY, US

	Brown University <i>Teaching Assistant</i> Course: Systems Transforming Systems Professor: Nikos Vasilakis Guided students through a systems research project, including implementation, evaluation, and paper writing. Paper submitted to USENIX ATC 2025.	Fall 2024 Providence, RI, US
	Brown University <i>Teaching Assistant</i> Course: Distributed Systems Professor: Nikos Vasilakis Helped revamp the course by redesigning its semester-long project and implementing its autograding infrastructure from scratch.	Spring 2024 Providence, RI, US
	National Technical University of Athens <i>Lab Assistant</i> Course: Operating Systems Professors: N. Koziris , G. Goumas	Spring 2022 Athens, Greece
	National Technical University of Athens <i>Lab Assistant</i> Course: Programming Techniques Professors: N. Papaspyrou , A. Pagourtzis , G. Stamou , G. Goumas	Spring 2020 Athens, Greece
	National Technical University of Athens <i>Lab Assistant</i> Course: Introduction to Programming Professors: N. Papaspyrou , D. Fotakis , S. Zachos	Fall 2019 Athens, Greece
Research Mentoring	Armaan Agrawal <i>Columbia University, BS</i>	May 2026 – Present
	Grace Li <i>Columbia University, BS</i>	Sep 2025 – Present
	Matthew Lou <i>Columbia University, BS → next: Hudson River Trading</i>	Sep 2025 – Jun 2026
	Joey Huang <i>Columbia University, BS → next: Google</i>	Sep 2025 – Dec 2025
Invited Talks	VineLM: Trie-Based Fine-Grained Control for Agentic Workflows Languages, Systems, and Data Seminar @ University of California, Santa Cruz Host: Lindsey Kuper	May 2025

Skills	Prog. Languages	Python, C, Shell, Javascript, Java, Go
	Frameworks	LangChain, LangGraph
	Platforms	AWS, Docker, Kubernetes, Apache Kafka, Apache Spark
	Tools	Git, LaTeX

Languages	English	Fluent — TOEFL iBT 116/120 (C2)
	Greek	Native